4.5 Scatter Plots and Lines of Fit

Name _____ Period ____ Date _____

Algebra I 4.5 · Student sheet

I can…

- interpret scatter plots and identify correlation.
- draw and write an equation for a line of fit.
- use a line of fit to make predictions.

Vocabulary · fill in the missing word

Scatter plot — a graph of _____ data as points

Positive / negative correlation — y tends to increase / decrease as x _____

No correlation — no clear _____

Line of fit — a line that _____ the trend of the data

Part 1 · Study hours

Hours	1	2	3	4	5	6
Score	62	68	71	79	83	88

Plot the data, draw a line of fit, and choose two points ON your line.

Correlation: _____ Equation: _____ Predicted score for 7 hours: _____

Part 2 · Notes: fill in the blanks

Positive correlation: as x increases, y tends to _____.

A line of fit is close to _____ of the points.

Correlation does not mean _____.

Part 3 · Examples (work them with the class)

Example 1

Tell whether each shows positive, negative, or no correlation.

a. age of a car and its value

b. height and shoe size

c. shoe size and test score

Example 2

Use the study-hours data. A line of fit passes through (1, 62) and (6, 88). Write its equation.

Study hours

Hours	1	2	3	4	5	6
Score	62	68	71	79	83	88

Example 3

Interpret the slope and y-intercept of $y = 5.2x + 56.8$, and predict the score for 7 hours.

Part 4 · Partner practice

Partner 1

Does the number of hours spent watching TV and the number of hours sleeping likely show positive, negative, or no correlation?

Partner 2

A line of fit for a plant’s height h (cm) after w weeks is h = 1.5w + 4. Predict the height after 8 weeks.